

Energy-efficient Analog Integrated Circuits

DCDC Converters – Assignment 2

Assignment

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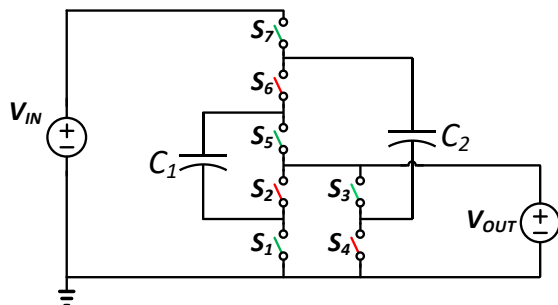
Introduction

In this assignment you will learn how to model and design two among the most popular Switched-Capacitor converter topologies and to understand what their benefits are for a certain set of design specifications. The first part of the Tutorial requires some hand calculation and a few plots of the derived equations, while the second part is done on cadence which will serve as a validation of the models as well as a design exercise.

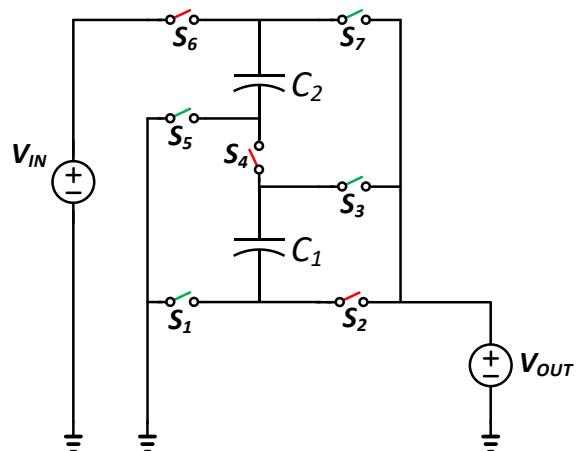
Part I – Modeling of the Dickson and Series-Parallel Switched-Capacitor Step-Down Converter

Given a Dickson and a Series-Parallel Switched-Capacitor topology with the same specifications as in the Tutorial:

Dickson



Series-Parallel



1. Draw circuit in phase 1 and phase 2
2. Calculate the voltage at which each flying capacitor will balance
3. Calculate the charge multiplier vectors for the slow switching limit R_{SSL}
4. Calculate the slow switching limit R_{SSL}
5. Calculate the charge multiplier vector for the fast switching limit R_{FSL}
6. Calculate the fast switching limit R_{FSL}
7. Calculate the converter Output Resistance
8. Write a comparison between the Ladder, Dickson and Series-parallel switched-capacitor converter topologies based on the following topics:
 - Conduction losses
 - Switching losses
 - Number of Flying Capacitors and Switches
 - Flying Capacitors and Switches utilization

Part II – Simulation of the Dickson Switched-Capacitor Converter on Cadence

1. Create the Dickson converter testbench taking inspiration from the ladder testbench used during the tutorial
 - a. Use ideal *vpulse* generators to control the switches and chose their swing to ensure a V_{gs} equal to 1.8V when the switch is closed and 0V when open, giving a 500mOhm $R_{DS(on)}$.
2. Run the Dickson converter testbench simulation and verify that the simulated output resistance matches the calculated one
3. Plot the converter time waveforms of the switching nodes and all the relevant signals
4. Simulate and calculate the converter output voltage
 - a. Choose an appropriate R_{load} range to observe a V_{OUT} drop down to 0.7V at the heaviest load
5. Calculate and Simulate the converter efficiency neglecting the gate switching losses
6. Calculate and Simulate the converter efficiency including the gate switching losses
7. Plot the simulated total conduction loss, the gate switching losses and their sum, all normalized to the output power
8. Can you suggest a design strategy to improve the converter efficiency?
9. Bonus Questions:
 - a. during all the previous analysis we have used all identical switches. However, as we are analysing step-down converters, when the voltage decreases the current increases. Therefore, we expect the current to be higher for the switches that are closer to the output. Could you devise a design strategy to optimize the converter efficiency sizing the switches accordingly to their current? Could you report the analysis and simulation results that motivate your strategy?